# Bihar Engineering University, Patna

## B. Tech 4th Semester Examination, 2024

*Course: B.Tech* | *Time: 03 Hours*
*Code: 100404* | *Subject: Discrete Mathematics* | *Full Marks: 70*

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*Instructions:-*
- *(i) The marks are indicated in the right-hand margin.*
- *(ii) There are NINE questions in this paper.*
- *(iii) Attempt FIVE questions in all.*
- *(iv) Question No. 1 is compulsory.*

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**Q.1** *Choose the correct option / answer the following (Any seven question only):-* [2 x 7 = 14]

(a) The function $f: R \rightarrow R$ defined by $f(x) = x^3 + 5$
(i) One-One but not onto
[✓ (ii) Onto but not One-One]
(iii) Both One-One and Onto
(iv) None of these

(b) Consider the following relation $R$ on a set $A = \{1,2,3,4\}$:
$R = \{(1,1),(1,3),(3,2),(2,2),(3,3),(3,1),\ (2,3),(1,4),(4,\ 1)\}$ Then
(i) Reflexive
(ii) Symmetric
(iii) Transitive
(iv) Equivalence

(c) Out of the following which of these integers is not prime?
(i) 23
(ii) 59
(iii) 21
(iv) 71

(d) The Greatest common divisor of 0 and 11
(i) 0
(ii) 1
(iii) 11
(iv) Does not exist

(e) Suppose G is a group with a binary operation '*' defined by $a * b = a + b + 1, \forall\ a, b \in G$ then
(i) Identity element = -1
(ii) Identity element = -2
(iii) Identity element = 1
(iv) Identity element = 2

(f) The field which contains at least _______ element/elements
(i) 0
[✓ (ii) 1]
(iii) 2
(iv) 3

(g) The set of integers with respect to addition is
(i) Abelian group
(ii) Cyclic group
(iii) Both (i) and (ii)
(iv) None of these

(h) The total number of edges in a complete graph of $n$ vertices is
(i) $n$
(ii) $\frac{n}{2}$
(iii) $n^2 - 1$
(iv) $\frac{n(n-1)}{2}$

(i) In any undirected graph, the sum of degrees of all vertices
(i) must be even
(ii) is a twice the number of edges
(iii) must be odd
(iv) Both (i) and (ii)

(j) $(p \rightarrow q) \wedge (r \rightarrow q)$ is equivalent to
(i) $(p \vee r) \rightarrow q$
(ii) $p \vee (r \rightarrow p)$
(iii) $p \vee (r \rightarrow q)$
(iv) $p \rightarrow (q \rightarrow r)$

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**Q.2** (a) Given A = {1, 2, 3}, B = {7, 8} and R = { (1, 7), (2, 7), (1, 8), (3, 8) }, find $R^{-1}$ (inverse of $R$ ) and $R'$ complement of $R$. [7]

(b) Consider $A = B = C = R$ set of real numbers and let $f: A \rightarrow B, g: B \rightarrow C$ defined by $f(x) = x + 9$ and $g(y) = y^2 + 3$, find the following composition functions: $(fog)(a), (gof)(a), (gof)(3), (fog)(-3)$. [7]

**Q.3** (a) Describe the Fundamental Theorem of Arithmetic. Factorize the number '324' and represent it as a product of primes. [7]

(b) Use the Euclidean algorithm to find the greatest common divisor of each pair of integers, (i) 60, 90 (ii) 414, 662 [7]

**Q.4** (a) What is the negation of each of the following propositions?
(i) Today is Tuesday (ii) A cow is an animal (iii) No one wants to buy my house. [7]

(b) Determine the truth value of each of the following statements (i) 4+3=7 and 6+2=8 (ii) 2+3=4 and 3+1=2. [7]

**Q.5** (a) Show that the set of rational numbers $Q$ forms an abelian group under the composition defined by $a * b = a + b - ab$. [7]

(b) Define a ring, give an example of a commutative ring without unity, and a non-commutative ring with identity. [7]

**Q.6** (a) Describe a graph model that can be used to represent all forms of electronic communication between two people in a single graph. What kind of graph is needed? [7]

(b) Explain the complete graph and prove that $k_5$ (complete graph) is a nonplanar graph. [7]

**Q.7** (a) Explain the congruence relation. Let $n$ be a positive integer then: (i) If $a \equiv b \ (mod\ n)$, then $b \equiv a \ (mod\ n)$ (ii) If $a \equiv b \ (mod\ n)$ and $b \equiv c \ (mod\ n)$, then $a \equiv c \ (mod\ n)$. [7]

(b) Show that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$. [7]

**Q.8** (a) What is the field? Show that $(Q, +, \times)$ is a field. [7]

(b) If in a ring $R$ with unity, $(xy)^2 = x^2y^2$ for all $x, y \in R$, then $R$ is commutative. [7]

**Q.9** (a) Write short notes of the following: (i) Eulerian Graph (ii) Graph Coloring [7]

(b) Write short notes of the following: (i) Chromatic number (ii) Perfect Graph [7]

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